Automatic Plant-Watering System for FabLab

A small automatic plant-watering system built using an Arduino, soil-moisture sensor, water pump, and relay. The system activates the pump when soil moisture drops below a threshold. Initial testing confirmed pump functionality, but sensor readings showed fluctuations. Optimal threshold and long-term reliability remain to be determined.

Objectives

Design and build a small automatic plant-watering system suitable for a FabLab environment. The system should monitor soil moisture levels and automatically activate a water pump when the soil becomes too dry, reducing the need for manual plant watering.

Context and Problem

Manual plant watering in a FabLab setting can be inconsistent or forgotten, potentially leading to plant stress or death. An automated solution provides a reliable way to maintain appropriate soil moisture levels without constant human intervention.

Materials and Tools

The following components were used in the system:

- Arduino microcontroller
- Soil-moisture sensor
- Water pump
- Relay module

Missing information:

- Specific Arduino model (e.g., Uno, Nano, Mega)
- Soil-moisture sensor model (e.g., capacitive vs. resistive type)
- Water pump specifications (voltage, flow rate, type)
- Relay module specifications (coil voltage, contact rating)
- Wiring diagram or schematic
- Power supply details for the system

Technical Choices

The system architecture uses a soil-moisture sensor to measure the moisture level in the plant pot. The Arduino reads the sensor values and compares them against a threshold. When the moisture level falls below the threshold (indicating dry soil), the Arduino activates the relay, which in turn powers the water pump to irrigate the plant.

Missing information:

- Threshold value for triggering the pump (not yet determined)
- Whether hysteresis or averaging is used to handle sensor fluctuations
- Code implementation details (e.g., polling interval, calibration method)

Procedure

1. Assemble the hardware components: connect the soil-moisture sensor, relay, and water pump to the Arduino according to the wiring configuration.

2. Upload the control program to the Arduino.

3. Place the soil-moisture sensor in the plant pot.

4. Test the system by observing sensor readings and pump activation.

Missing information:

- Detailed wiring connections (pin assignments for sensor, relay, pump)
- Arduino code or pseudocode
- Calibration procedure for the sensor

Difficulties and Solutions

During testing, the sensor readings were observed to fluctuate, which could lead to inconsistent pump activation. This is a common issue with soil-moisture sensors and may require software filtering (e.g., averaging multiple readings) or hardware improvements (e.g., using a more stable capacitive sensor instead of resistive).

Missing information:

- Whether any mitigation strategies were attempted for the sensor fluctuations
- Root cause analysis of the fluctuations (e.g., electrical noise, sensor degradation, poor contact)

Results and Tests

The system was tested with a plant in a pot. The water pump operated correctly when activated, confirming that the pump and relay function as expected. However, the sensor readings exhibited fluctuations, and the optimal moisture threshold for triggering the pump has not yet been determined.

Missing information:

- Quantitative sensor readings (e.g., typical values for dry vs. moist soil)
- Duration of pump activation per watering cycle
- Number of watering cycles tested
- Plant species used in testing

Possible Improvements

1. Determine the optimal moisture threshold through systematic testing.

2. Implement sensor data smoothing (e.g., moving average) to reduce reading fluctuations.

3. Conduct long-term testing over several weeks to evaluate system reliability.

4. Consider adding a water reservoir with level monitoring to prevent pump dry-running.

5. Add user feedback (e.g., LED indicator, serial output) to monitor system status.

Missing information:

- Plans for long-term testing (timeline, monitoring method)
- Whether additional features are planned (e.g., manual override, notifications)

Missing Information

The following information is not specified in the project description and would be valuable for reproducibility:

- Specific models and specifications of all components (Arduino, sensor, pump, relay)
- Wiring diagram or pin assignments
- Arduino code or algorithm details
- Calibration and threshold determination methodology
- Sensor fluctuation analysis and mitigation strategies
- Long-term testing plan and results
- Power supply configuration
- Water reservoir or source details